

Extending the Renewable-Resource Fishery Model:

Adding a Substitute-Good Market (Chicken) to Meadows' Fig. 42

1. Background and Motivation

Donella Meadows' fishery example in *Thinking in Systems* (Fig. 42) models a capital stock (fishing fleet) that grows through a reinforcing investment loop and is constrained by a renewable resource stock (fish) through a balancing loop: more capital drives more harvest, which depletes the resource, which lowers the yield per unit of capital, which lowers profit and slows investment. Depending on the strength of that feedback relative to the resource's regeneration curve, the book shows three qualitative outcomes: smooth adjustment to equilibrium, oscillation, or overshoot and collapse.

This model extends that structure with one additional mechanism: a **substitute-good market**. A second stock, *Chicken Meat Stock*, is produced and consumed independently of the fishery, at a fixed base rate. A *Chicken Demand Ratio* converter reads the fish *Price* and returns a value between 0 and 1 that grows as fish becomes more expensive — representing consumers switching toward chicken as fish gets pricier. That ratio then reduces the *Harvest* flow directly: $\text{Harvest} = \text{Capital} \times \text{Yield Per Unit Capital} \times (1 - \text{Chicken Demand Ratio})$. The idea being tested is whether a market-level substitution effect can act as a second balancing loop on the fishery, changing whether the system settles, oscillates, or collapses.

2. Model Structure

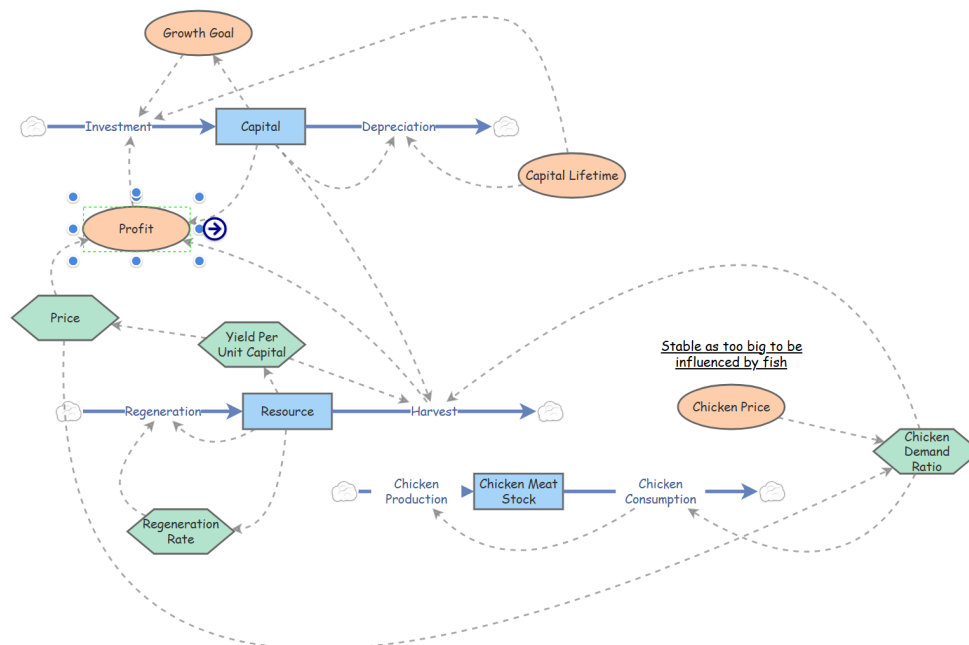


Figure 1. Stock-and-flow diagram. The core fishery loops (top and left) follow Meadows' Fig. 42: a reinforcing loop of Investment building Capital, and a balancing loop of Harvest depleting Resource, moderated by Yield Per Unit Capital and Regeneration Rate. The added structure (right) is the chicken sub-market: Chicken Production and Consumption keep Chicken Meat Stock roughly steady, while Chicken Demand Ratio — driven by fish Price — feeds back into the fishery's Harvest equation.

2.1 Key relationships

Element	Definition
Investment	$\text{Max}(0, \text{Min}(\text{Profit}, \text{Capital} \times \text{Growth Goal} + \text{Capital} / 20))$
Depreciation	$\text{Capital} / \text{Capital Lifetime}$ (Capital Lifetime = 20 years)
Profit	$(\text{Harvest} \times \text{Price}) - (\text{Capital} \times 0.1)$
Harvest	$\text{Capital} \times \text{Yield Per Unit Capital} \times (1 - \text{Chicken Demand Ratio})$
Regeneration	$\text{Resource} \times \text{Regeneration Rate}$
Yield Per Unit Capital	Graphical fn. of Resource (rises from 0 to 0.35 as Resource: 0→1000)
Regeneration Rate	Graphical fn. of Resource (hump-shaped: peaks near Resource = 500)
Price	Graphical fn. of Yield Per Unit Capital (falls from 10 to 1 as yield rises)
Chicken Demand Ratio	Graphical fn. of Price (rises from 0 to 1 as fish Price: 0→3)
Chicken Consumption / Production	$100 \times (1 + \text{Chicken Demand Ratio})$
Growth Goal (baseline)	5% per year
Initial Capital / Resource	10 / 1000

3. Baseline Run (Chicken Price fixed at 3, Growth Goal = 5%)

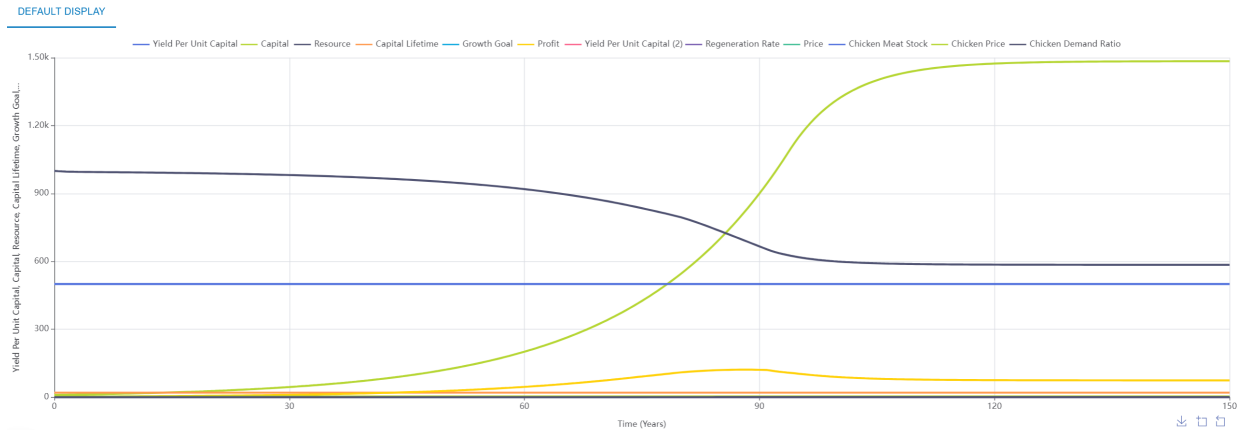


Figure 2. Baseline simulation, 150 years. Capital (olive) and Resource (dark grey) are the two dominant curves; the remaining variables are small enough on this shared axis to sit near zero.

Resource declines gradually from 1000 to a stable plateau of roughly 580–590, reached by about year 100. Capital grows in an S-shape, with its steepest climb between years 60 and 100, and levels off near 1480–1490. Profit (not very visible on this scale but confirmed in the run) rises to a peak of about 120–130 around year 85–90 and then eases back to a steady value near 70 once Capital's growth slows. None of the runs at these settings show oscillation or collapse: the system moves smoothly to a single equilibrium, closer to Meadows' Figure 43 ("overshoot and adjustment to equilibrium") than to the oscillating or collapsing cases in Figures 44–45.

4. Sensitivity Test: Growth Goal Raised to 11%

This run keeps Chicken Price at the baseline value of 3 but more than doubles the industry's target growth rate, from 5% to 11% per year, to test whether a more aggressive reinvestment policy is enough to overwhelm the chicken-substitution balancing loop.

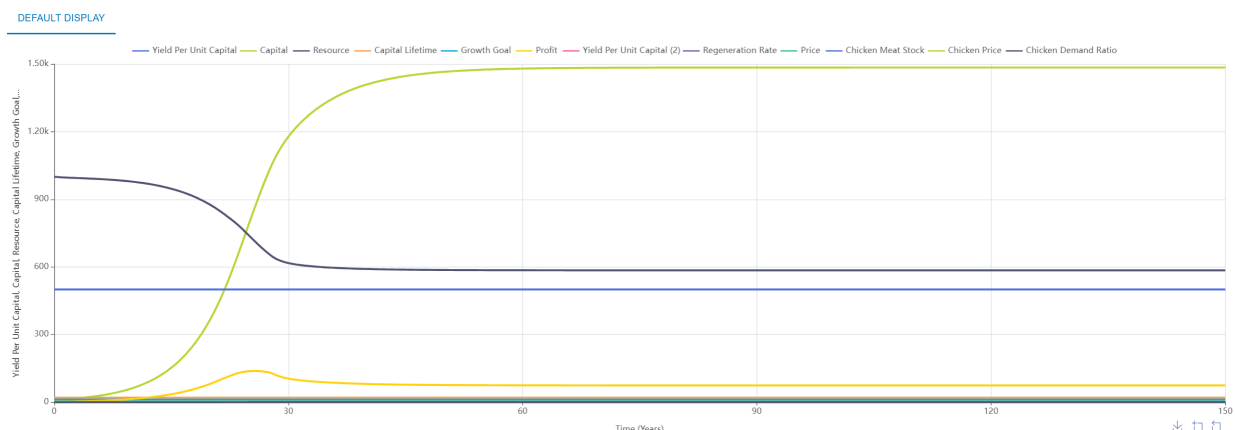


Figure 4. Simulation with Growth Goal = 11% (all else at baseline).

The qualitative outcome is the same as the baseline — Resource still settles near 580–590 and Capital still plateaus near 1480–1490, with Profit still peaking and then relaxing to a steady value — but the timing compresses sharply. Capital's steep climb, which took from about year 60 to year 100 at 5% growth, now happens between roughly year 10 and year 50, and Resource has already fallen to its floor by around year 30–40 instead of year 100. The system still finds the same equilibrium; it just gets there about two to three times faster. This matches the book's discussion of the nonrenewable-resource case (Fig. 40): raising the reinvestment/growth rate does not change where the system ends up, but it does compress the timeline — the resource is drawn down and the industry matures much sooner.

5. Control Test: Removing the Chicken-Substitution Loop

Sections 3 and 4 show that the model, with the chicken-substitution mechanism active, always settles into a stable equilibrium rather than oscillating or collapsing. That is only meaningful if the same model, at the same parameter values, behaves differently once that mechanism is switched off. The runs below repeat the baseline and the high-growth scenario with the Harvest term's $(1 - \text{Chicken Demand Ratio})$ reduction disabled, isolating what the chicken loop was actually doing.

5.1 Same growth rate (5%), chicken loop removed



Figure 5. Growth Goal = 5%, chicken-substitution loop disabled (compare directly to the baseline, Figure 2).

This is the key comparison. With every other setting identical to the baseline, removing the chicken loop turns a smooth convergence into an overshoot-and-collapse cycle: Capital overshoots to roughly 1350 around year 85 (versus a smooth plateau near 1480 with the loop active), Resource is driven down toward zero by around year 90–100, and Profit swings from a peak of about 200 down to roughly –50 before slowly recovering. Capital then falls sharply from its peak and only begins a slow, partial rebound by year 150. In other words, at exactly the parameter values where the full model settles quietly, the same model without the substitution loop overshoots the resource's carrying capacity and partially collapses — the pattern Meadows describes in Figures 44–45. This is direct evidence that the chicken-substitution loop, not some other feature of the parameters chosen, is what stabilizes the baseline run.

5.2 High growth rate (11%), chicken loop removed

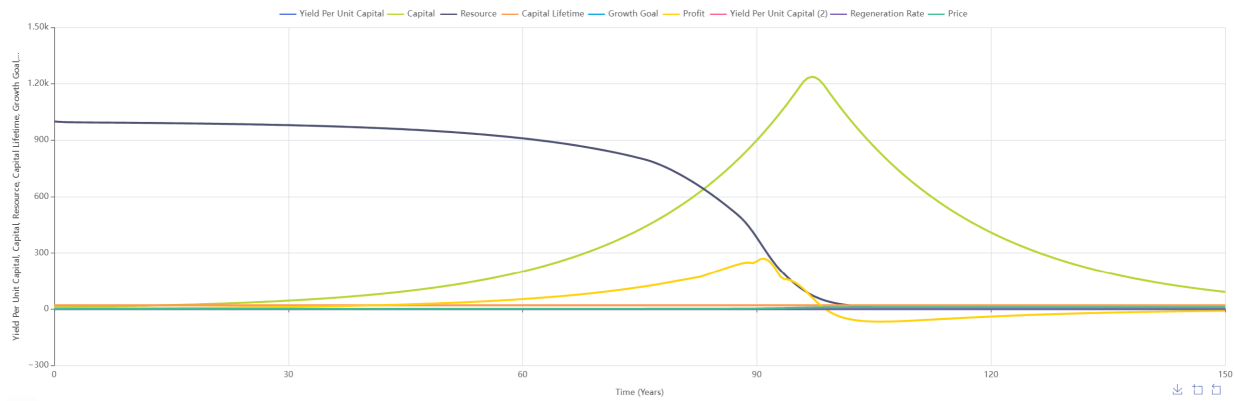


Figure 6. Growth Goal = 11%, chicken-substitution loop disabled (compare to Figure 4, the same growth rate with the loop active).

Removing the loop at the higher growth rate makes the instability more severe, not less: Capital overshoots further, to roughly 1240 by year 95, while Resource collapses almost completely by around year 100. Profit follows the same overshoot-and-undershoot pattern as in 5.1, dipping to about -100 before beginning a slow recovery, and Capital itself falls to under 100 by year 150 with no meaningful rebound in this window. With the chicken loop active at the same 11% growth rate (Figure 4), the system instead converges smoothly to the same ~1480/~585 equilibrium as the baseline, just faster. The contrast between Figures 4 and 6 is the clearest demonstration in this report that the substitution loop is doing real stabilizing work, and that its effect gets more important, not less, as the reinvestment policy becomes more aggressive.

5.3 Growth Goal set arbitrarily high (Growth Goal = 1), chicken loop removed



Figure 7. Growth Goal = 1 (i.e., a 100% annual growth target, effectively removing the growth-rate constraint), chicken-substitution loop disabled.

This run tests what happens when the growth-rate target itself stops being the limiting factor. Even with Growth Goal set absurdly high, Capital does not run away: it rises quickly and levels off at only about 230, and Resource declines only mildly, to roughly 900. The reason is visible in the Investment equation,

$\text{Investment} = \text{Max}(0, \text{Min}(\text{Profit}, \text{Capital} \times \text{Growth Goal} + \text{Capital} / 20))$: once Growth Goal is large enough, the $\text{Min}(\text{Profit}, \dots)$ term takes over, and Investment is capped by whatever Profit is actually available rather than by the growth target. So this particular extreme case is governed by the model's investment-financing constraint, not by the chicken loop or the resource-depletion loop — a useful reminder that Growth Goal only matters as a lever up to the point where profit becomes the binding constraint instead.

6. Discussion

Sections 3 and 4 showed that, with the chicken-substitution loop active, both the baseline growth rate and a much higher one converge smoothly to essentially the same equilibrium. Section 5's control runs show that this is not simply a property of the parameter values chosen: with the identical parameters but the $(1 - \text{Chicken Demand Ratio})$ term removed from Harvest, the same model overshoots and partially collapses, and does so more severely as Growth Goal increases (Figures 5–6). That pairing — same settings, opposite qualitative outcome depending only on whether the substitution loop is present — is the strongest evidence in this report that the loop is doing real stabilizing work, not just producing similar-looking numbers coincidentally. Because Harvest is scaled down by $(1 - \text{Chicken Demand Ratio})$ as fish Price climbs, the full model effectively has two balancing loops acting on Capital's growth instead of one: the original resource-depletion loop (falling yield $\rightarrow$ falling profit $\rightarrow$ less investment) and the market-substitution loop (rising price $\rightarrow$ more chicken demand $\rightarrow$ less harvest allowed $\rightarrow$ less profit). With both loops active, the balancing side of the system dominates soon enough that Resource never drops below the threshold where it would fail to regenerate.

One limitation worth noting: an earlier test that lowered Chicken Price from 3 to 2 produced almost no change in the fishery's trajectory. That is because Chicken Demand Ratio, as built, is a graphical function of fish Price alone and never reads Chicken Price at all — so Chicken Price affects only the chicken sub-market's own bookkeeping, not the feedback loop that protects the fishery. A more realistic version would make Chicken Demand Ratio a function of the price ratio ($\text{Price} / \text{Chicken Price}$) instead, so that a cheaper substitute would actually accelerate the switch away from fish.

7. Conclusion

Adding a substitute-good market to Meadows' fishery model introduces a second balancing loop that stabilizes the system. With the loop active, Capital and Resource converge to the same equilibrium (~1480 and ~585) whether Growth Goal is 5% or 11% — only the speed of convergence changes. With the loop removed, the identical parameter sets instead produce overshoot and partial collapse, and the collapse is worse at the higher growth rate. An additional control run showed that setting Growth Goal arbitrarily high does not itself cause runaway growth, because Investment is capped by available Profit regardless of the growth target. The main limitation identified is that the substitution mechanism reads absolute fish price rather than fish price relative to chicken price, which is why varying Chicken Price alone had little effect and is the clearest direction for future refinement.